

SAMPREETH AVVARI

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SUMMARY

Machine Learning Engineer who designs, trains, and ships models across the full lifecycle, from data pipelines through serving and monitoring. Built a CT-scan classifier and a Llama RLHF pipeline (67% win rate) among 5 production systems, solo.

EDUCATION

MS, Computer Engineering, New York University, NY (CGPA 3.84/4)

Aug 2023 - May 2025

BTech, Computer Science and Engineering, JNTU, India (CGPA 3.92/4)

Jul 2017 - Jun 2021

SKILLS

ML / Modeling: PyTorch, Transformers, Hugging Face, TRL, fine-tuning, RLHF (GRPO/PPO), QLoRA, reward models, nnU-Net, classification, ranking, embeddings, evaluation (holdouts, A/B, AUROC, win-rate), experiment tracking (WandB, MLflow)

MLOps & Data: MLflow, Airflow, Ray, Docker, Kubernetes, Terraform, CI/CD, model serving (OpenVINO, Cloud Run), monitoring (Prometheus, Grafana, OpenTelemetry); Spark, Kafka, Airflow ETL, SQL, BigQuery, PostgreSQL, pgvector, feature pipelines

Languages & SWE: Python, Go, Java, C++, TypeScript, SQL; FastAPI, REST, microservices, Pydantic, pytest, code review

Applied AI / LLM: LLM pipelines, prompt engineering, JSON Schema contracts, RAG (pgvector, Milvus), vector search, Vertex AI Gemini, Llama, guardrails; GCP, AWS, Google Gen AI Leader

WORK EXPERIENCE

Machine Learning Engineer | Hybridge Implants, New York

Sep 2025 - Present

- Partnered with clinicians and the C-suite to ship 5 production ML and LLM systems solo, owning the full lifecycle from data and training to serving and monitoring.

CBCT Scan Validator, in-house dental-CT quality classifier

- Built a CT scan-quality classifier on a frozen nnU-Net v2 encoder with a compact 500K-parameter head, served via OpenVINO at 5.5s per scan on CPU-only Cloud Run.
- Owned the full MLOps loop (SFTP ingest, parity-checked ETL, training, serving) and gated releases on a 20-scan holdout that caught a leaky 0.80 AUROC, shipping an honest 0.63.

Doc Coach, LLM consultation-QA pipeline

- Shipped an LLM consultation-QA pipeline (Vertex AI Gemini) scoring every consult on a 7-criterion rubric, with versioned prompts and JSON Schema guardrails cutting hallucinations 35%.
- Raised treatment acceptance 130% and downstream revenue 43% by iterating the rubric and prompts against labeled evaluations.

Treatment Estimator, Next.js and Postgres quoting platform

- Delivered a 5-model pricing engine behind a Next.js and Postgres platform serving 50 consults a week, replacing a decade-old vendor tool.
- Built pricing and eligibility features from messy clinical data, with Postgres-enforced invariants and 100% branch test coverage.

Software Engineer | Shure Incorporated, India

Aug 2021 - Aug 2023

- Engineered event-driven Flask REST APIs on AWS ingesting 500K+ daily Kafka events in real time, cutting processing time 20% with async workers.
- Directed delivery across 3 release tracks with Jenkins canary deploys, cutting deploy errors 40% with 150+ pytest tests at 90%+ coverage, and led code reviews.

RESEARCH EXPERIENCE

Machine Learning Researcher | New York University (writeup)

May 2024 - Sep 2025

- Fine-tuned Llama 3.1 8B through a full RLHF lifecycle (SFT then GRPO/PPO) training QLoRA reward models, with WandB experiment tracking and a 67% win rate on human evaluation.
- Engineered a Python and Airflow ETL turning 118 Reddit CMV shards into 38K preference pairs with schema enforcement, failure alerting, and checkpointed recovery.

PROJECTS

JobPilot, open-source (MIT) ML-driven job-hunt autopilot

- Architected JobPilot, an open-source job-hunt autopilot (12 users) scoring candidate-job fit with an LLM-as-judge model and auto-tailoring resumes across 11 sources for under \$10 a month.
- Designed a plug-in source architecture (11 ATS connectors behind one schema), a self-refining rewrite loop, and a Sheets-as-database, shipped with CI/CD and 50+ tests.

Loan Radar, self-retraining loan-default scoring platform

- Packaged Loan Radar, a self-retraining loan-default scoring model on Terraform Kubernetes (Ray) with automated retraining and MLflow-tracked CI/CD for rollback-safe serving across 3 microservices.